

SOLUTION

RESOLUTION ALGORITHM IMPLEMENTATION

Question 1 – Rain and Wet Ground

1. Propositional Logic

Let:

- **R** = It is raining.
- **W** = Ground is wet.

Knowledge Base:

1. If it rains, the ground becomes wet.
 $R \rightarrow W$
2. It is raining.
R

2. Convert into CNF

The implication:

$$R \rightarrow W$$

is converted as:

$$\neg R \vee W$$

Therefore, the clauses are:

- **C1: $\neg R \vee W$**
- **C2: R**

3. Negate the Goal

Goal: **W**

Negated goal:

$$\neg W$$

Add it to the clauses:

- **C1: $\neg R \vee W$**
- **C2: R**
- **C3: $\neg W$**

4. Apply Resolution

Resolve C1 and C3:

$$(\neg R \vee W), \neg W$$

Therefore:

$$\neg R$$

Now resolve:

$$R, \neg R$$

Therefore:

□ (Empty Clause)

5. Conclusion

Since the **empty clause** (□) is obtained, the negation of the goal leads to a contradiction.

Therefore, the ground is wet.

Question 2 – Student Assignment Submission

1. Propositional Logic

Let:

- **S** = Rahul submitted the assignment.
- **M** = Rahul receives internal marks.

Knowledge Base:

1. If Rahul submits the assignment, Rahul receives internal marks.
 $S \rightarrow M$
2. Rahul submitted the assignment.
S

2. CNF Conversion

$$S \rightarrow M$$

becomes:

$$\neg S \vee M$$

Clauses:

- **C1: $\neg S \vee M$**
- **C2: S**

3. Negate the Goal

Goal:

M

Negated goal:

$\neg M$

Clause set:

- C1: $\neg S \vee M$
- C2: S
- C3: $\neg M$

4. Resolution

Resolve C1 and C3:

$(\neg S \vee M), \neg M$

Result:

$\neg S$

Now resolve C2 and the new clause:

S, $\neg S$

Result:

□

5. Conclusion

The empty clause is obtained.

Therefore, Rahul receives internal marks.

Question 3 – Library Membership

1. Propositional Logic

Let:

- **L** = Priya is a library member.
- **B** = Priya can borrow books.

Knowledge Base:

1. If Priya is a library member, she can borrow books.
 $L \rightarrow B$
2. Priya is a library member.
L

2. CNF Conversion

$L \rightarrow B$

becomes:

$\neg L \vee B$

Clauses:

- **C1: $\neg L \vee B$**
- **C2: L**

3. Negate the Goal

Goal:

B

Negated goal:

$\neg B$

Clauses:

- **C1: $\neg L \vee B$**
- **C2: L**
- **C3: $\neg B$**

4. Resolution

Resolve C1 and C3:

$(\neg L \vee B), \neg B$

Result:

$\neg L$

Now resolve C2 and the new clause:

$L, \neg L$

Result:

□

5. Conclusion

The empty clause is derived.

Therefore, Priya can borrow books.

Question 4 – Placement Eligibility

1. Propositional Logic

Let:

- A = Arun cleared the aptitude test.
- P = Arun is eligible for placement.

Knowledge Base:

1. If Arun clears the aptitude test, he is eligible for placement.

$A \rightarrow P$

2. Arun cleared the aptitude test.

A

2. CNF Clauses

Convert:

$A \rightarrow P$

into:

$\neg A \vee P$

Therefore:

- **C1: $\neg A \vee P$**
- **C2: A**

3. Negate the Goal

Goal:

P

Negated goal:

$\neg P$

Clause set:

- C1: $\neg A \vee P$
- C2: A
- C3: $\neg P$

4. Resolution Step-by-Step

Resolve:

C1: $\neg A \vee P$

with

C3: $\neg P$

Result:

$\neg A$

Now resolve:

C2: A

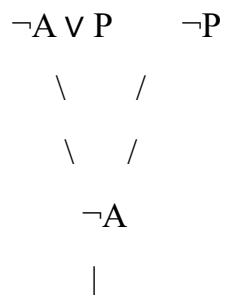
with:

$\neg A$

Result:

□

5. Resolution Tree



| A

|

□

6. Conclusion

Since the empty clause is obtained, the negated goal $\neg P$ is contradictory.

Therefore, Arun is eligible for placement.

Question 5 – Access Control System

1. Propositional Logic

Let:

- **C** = User entered the correct password.
- **A** = User is authenticated.
- **G** = User is granted access.

Knowledge Base:

1. If the user enters the correct password, the user is authenticated.
 $C \rightarrow A$
2. If the user is authenticated, the user is granted access.
 $A \rightarrow G$
3. The user entered the correct password.
C

2. Convert into CNF

First implication:

$$C \rightarrow A$$

becomes:

$$\neg C \vee A$$

Second implication:

$$A \rightarrow G$$

becomes:

$\neg A \vee G$

Therefore:

- **C1:** $\neg C \vee A$
- **C2:** $\neg A \vee G$
- **C3:** C

3. Negate the Goal

Goal:

G

Negated goal:

$\neg G$

Complete clause set:

- **C1:** $\neg C \vee A$
- **C2:** $\neg A \vee G$
- **C3:** C
- **C4:** $\neg G$

4. Apply Resolution

First resolve:

C2: $\neg A \vee G$

with

C4: $\neg G$

Result:

$\neg A$

Next resolve:

C1: $\neg C \vee A$

with:

$\neg A$

Result:

$\neg C$

Now resolve:

C3: C

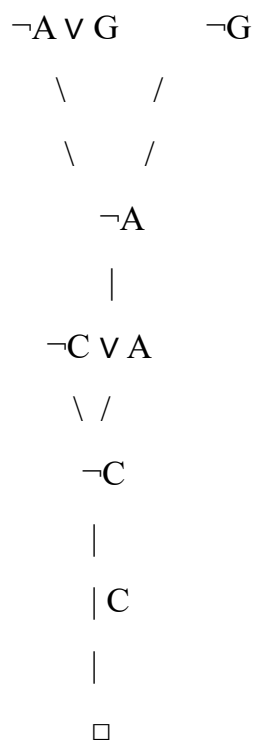
with:

$\neg C$

Result:

□

5. Resolution Tree



6. Conclusion

The empty clause □ is obtained. Therefore, the negation of the goal is false, proving the original goal.

Therefore, the user is granted access.